链式法则1

设 $u=\varphi(x), v=\psi(x)$ 在点 x 处可导，函数 $z=f(u,v)$ 在对应点 (u,v) 处可微，则

$z=f(\varphi(x),\psi(x))$ 在 x 处可导. 有

全导数 $\quad \dfrac{dz}{dx}=\dfrac{\partial z}{\partial u}\cdot\dfrac{du}{dx}+\dfrac{\partial z}{\partial v}\cdot\dfrac{dv}{dx}$

$z=f(u,v)$ 可微

$\Rightarrow \Delta z=\dfrac{\partial z}{\partial u}\Delta u+\dfrac{\partial z}{\partial v}\Delta v+o(\rho) \qquad \rho=\sqrt{\Delta u^2+\Delta v^2}$

$\Rightarrow \dfrac{\Delta z}{\Delta x}=\dfrac{\partial z}{\partial u}\cdot\dfrac{\Delta u}{\Delta x}+\dfrac{\partial z}{\partial v}\dfrac{\Delta v}{\Delta x}+\dfrac{o(\rho)}{\Delta x}$

$\Delta x\to 0$ 时 $\quad u=\varphi(x) \quad v=\psi(x)$ 可导.

$\Rightarrow \lim\limits_{\Delta x\to 0}\dfrac{\Delta u}{\Delta x}=\dfrac{du}{dx} \qquad \lim\limits_{\Delta x\to 0}\dfrac{\Delta v}{\Delta x}=\dfrac{dv}{dx}$

$\lim\limits_{\Delta x\to 0}\dfrac{o(\rho)}{\Delta x}=\lim\limits_{\Delta x\to 0}\dfrac{o(\rho)}{\rho}\cdot\sqrt{\left(\dfrac{\Delta u}{\Delta x}\right)^2+\left(\dfrac{\Delta v}{\Delta x}\right)^2}=0$

故 $\dfrac{dz}{dx}=\lim\limits_{\Delta x\to 0}\dfrac{\Delta z}{\Delta x}=\dfrac{\partial z}{\partial u}\dfrac{du}{dx}+\dfrac{\partial z}{\partial v}\dfrac{dv}{dx}$

例: $y = e^{uv} + u^2$, $u = \sin x$, $v = \cos x$ 求全导数 $\frac{dy}{dx}$

$$\frac{\partial y}{\partial u} = v\cdot e^{uv} + 2u \qquad \frac{\partial y}{\partial v} = u\cdot e^{uv}$$

$$\frac{du}{dx} = \cos x \qquad \frac{dv}{dx} = -\sin x$$

$$\Rightarrow \frac{dy}{dx} = \cos^2 x\left(e^{\sin x\cos x} + 2\sin x\right) - \sin^2 x\cdot e^{\sin x\cos x}$$

$$= e^{\sin x\cos x}\cdot\cos 2x + \sin 2x$$

链式法则2

设 $u=\varphi(x,y)$, $v=\psi(x,y)$ 在点 (x,y) 处可偏导，函数 $z=f(u,v)$ 在 (u,v) 处可微，则复合函数 $z=f(\varphi(x,y),\psi(x,y))$ 在点 (x,y) 处可偏导. 有公式

$$\frac{\partial z}{\partial x} = \frac{\partial z}{\partial u}\cdot\frac{\partial u}{\partial x} + \frac{\partial z}{\partial v}\cdot\frac{\partial v}{\partial x}$$

$$\frac{\partial z}{\partial y} = \frac{\partial z}{\partial u}\cdot\frac{\partial u}{\partial x} + \frac{\partial z}{\partial v}\cdot\frac{\partial v}{\partial x}$$

例：$z=u^{v}$, $u=x^2+y^2$, $v=xy$. 求 $\frac{\partial z}{\partial x}$, $\frac{\partial z}{\partial y}$

$$\frac{\partial z}{\partial x}=\frac{\partial z}{\partial u}\cdot\frac{\partial u}{\partial x}+\frac{\partial z}{\partial v}\cdot\frac{\partial v}{\partial x}$$

$$=v\cdot u^{v-1}\cdot 2x+\ln u\cdot u^{v}\cdot y$$

$$=(2xv+uy\cdot\ln u)\,u^{v-1}$$

$$=[2x^2y+y(x^2+y^2)\cdot\ln(x^2+y^2)]\cdot(x^2+y^2)^{xy-1}$$

$$\frac{\partial z}{\partial y}=\frac{\partial z}{\partial u}\cdot\frac{\partial u}{\partial y}+\frac{\partial z}{\partial v}\cdot\frac{\partial v}{\partial y}$$

$$=v\cdot u^{v-1}\cdot 2y+\ln u\cdot u^{v}\cdot x$$

$$=(2yv+ux\cdot\ln u)\,u^{v-1}$$

$$=[2xy^2+x(x^2+y^2)\ln(x^2+y^2)]\cdot(x^2+y^2)^{xy-1}$$

链式法则3

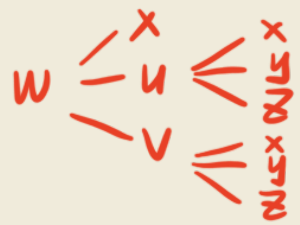
设 $u=\varphi(x,y)$, $v=\psi(x,y)$ 在 (x,y) 处可偏导. 函数 $z=f(x,y,u,v)$ 在点 (x,y,u,v) 处可微，则复合函数 $z(x,y)=f(x,y,\varphi(x,y),\psi(x,y))$ 在点 (x,y) 处可偏导.

$$\frac{\partial z}{\partial x}=\frac{\partial f}{\partial x}+\frac{\partial f}{\partial u}\cdot\frac{\partial u}{\partial x}+\frac{\partial f}{\partial v}\cdot\frac{\partial v}{\partial x}$$

(∂f 不可写作 ∂z)

$$\frac{\partial z}{\partial y}=\frac{\partial f}{\partial y}+\frac{\partial f}{\partial u}\cdot\frac{\partial u}{\partial y}+\frac{\partial f}{\partial v}\cdot\frac{\partial v}{\partial x}$$

例： $w=f(x,u,v)$ 可微 $u=x^2+xy+yz$，$v=xyz$，求 $\frac{\partial w}{\partial x}$，$\frac{\partial w}{\partial y}$，$\frac{\partial w}{\partial z}$



$$u=x^2+xy+\frac{v}{x}$$

$$\frac{\partial w}{\partial x}=\frac{\partial f}{\partial x}+\frac{\partial f}{\partial u}\cdot\frac{\partial u}{\partial x}+\frac{\partial f}{\partial v}\cdot\frac{\partial v}{\partial x}=f'_x+(2x+y)f'_u+yzf'_v$$

$$\frac{\partial w}{\partial y}=\frac{\partial f}{\partial u}\cdot\frac{\partial u}{\partial y}+\frac{\partial f}{\partial v}\cdot\frac{\partial v}{\partial y}=(x+z)f'_u+xzf'_v$$

$$\frac{\partial w}{\partial z}=\frac{\partial f}{\partial u}\cdot\frac{\partial u}{\partial z}+\frac{\partial f}{\partial v}\cdot\frac{\partial v}{\partial z}=yf'_u+xyf'_v$$

链式法则4

$z=f(u_1,u_2,\dots u_m)$ 可微，$u_j=g_j(x_1,x_2,\dots x_n)$ 可偏导.

$z=z(x_1,x_2,\dots x_n)$ 可偏导. 且

$$\frac{\partial z}{\partial x_i}=\sum_{j=1}^{m}\frac{\partial f}{\partial u_j}\cdot\frac{\partial g_j}{\partial x_i}$$

一阶全微分形式不变性

$$dz=\frac{\partial z}{\partial u}du+\frac{\partial z}{\partial v}dv=\frac{\partial z}{\partial x}dx+\frac{\partial z}{\partial y}dy$$

例: $z=e^u+\sin v$, $u=x+y$, $v=xy$, 求 $\frac{\partial z}{\partial x}$, $\frac{\partial z}{\partial y}$

$$dz=d(e^u+\sin v)=e^u du+\cos v\, dv$$

而 $du=dx+dy$ $\quad dv=x dy+y dx$

$$\Rightarrow dz=e^{x+y}dx+e^{x+y}dy+\cos xy\cdot(x dy+y dx)$$

$$\Rightarrow \frac{\partial z}{\partial x}=e^{x+y}+y\cdot\cos xy \qquad \frac{\partial z}{\partial y}=e^{x+y}+x\cos xy$$

复合函数的高阶偏导数

设 $z=f(x+y,xy,\frac{x}{y})$, f 的二阶偏导数连续, 试求 $\frac{\partial^2 z}{\partial x\partial y}$

$$\frac{\partial z}{\partial x}=f_1'+y\cdot f_2'+\frac{1}{y}f_3'$$

而 f_1', f_2', f_3' 仍是 x,y 的复合函数

$$\frac{\partial^2 z}{\partial x\partial y}=\frac{\partial f_1'}{\partial y}+f_2'+y\frac{\partial f_2'}{\partial y}-\frac{1}{y^2}f_3'+\frac{1}{y}\frac{\partial f_3'}{\partial y}$$

$$=1f_{11}''+xf_{12}''-\frac{x}{y^2}f_{13}''+f_2'+y(f_{21}''+xf_{22}''-\frac{x}{y^2}f_{23}''-\frac{1}{y^2}f_3'+\frac{1}{y}(f_{31}''+xf_{32}''-\frac{x}{y^2}f_{33}'')$$

$$=f_2'-\frac{1}{y^2}f_3'+f_{11}''+xyf_{22}''-\frac{x}{y^3}f_{33}''+(x+y)f_{12}''+(\frac{1}{y}-\frac{x}{y^2})f_{13}''$$

隐函数求导

$F(x,f(x))=0$ $\quad F\begin{cases} x \\ y-x \end{cases}$

$F'_x+F'_y\cdot f'(x)=0$

若 $F'_y\neq 0$，则 $f'(x)=-\dfrac{F'_x}{F'_y}$

函数 $z=F(x,y)$ 空间曲面

曲面 $z=F(x,y)$ 与 $z=0$ 相交一定为一条曲线

例. $1+y+\sin(x^2+y^2)=e^{xy}$ 在 $(0,0)$ 点

$F(x,y)=1+y+\sin(x^2+y^2)-e^{xy}$

1) $F(x,y)$ 连续可微

2) $F(0,0)=0$

3) $\dfrac{\partial F}{\partial y}(0,0)=1+2y\cdot\cos y^2\neq 0$

$$y'(x)=-\frac{F'_x}{F'_y}=\frac{2x\cdot\cos(x^2+y^2)-ye^{xy}}{1+2y\cdot\cos(x^2+y^2)-xe^{xy}}$$

$F(x,y,f(x,y))=0$

$$\frac{\partial F}{\partial x}+\frac{\partial F}{\partial z}\cdot\frac{\partial f}{\partial x}=0 \Rightarrow \frac{\partial f}{\partial x}=-\frac{\frac{\partial F}{\partial x}}{\frac{\partial F}{\partial z}}$$

$$\frac{\partial F}{\partial y}+\frac{\partial F}{\partial z}\cdot\frac{\partial f}{\partial y}=0 \Rightarrow \frac{\partial f}{\partial y}=-\frac{\frac{\partial F}{\partial y}}{\frac{\partial F}{\partial z}}$$

例：设 $u+e^u=xy$. 确定 $u=u(x,y)$ 求 $\frac{\partial^2 u}{\partial x\partial y}$

$F(x,y,u)=xy-u-e^u$

$$\frac{\partial u}{\partial x}=-\frac{\frac{\partial F}{\partial x}}{\frac{\partial F}{\partial u}}=-\frac{y}{-1-e^u}=\frac{y}{1+e^u} \qquad \frac{\partial u}{\partial y}=-\frac{\frac{\partial F}{\partial y}}{\frac{\partial F}{\partial u}}=\frac{x}{1+e^u}$$

$$\frac{\partial^2 u}{\partial x\partial y}=\frac{1}{1+e^u}-\frac{y\cdot e^u\cdot\frac{x}{1+e^u}}{(1+e^u)^2}$$

$$=\frac{1}{1+e^u}-\frac{y\cdot e^u x}{(1+e^u)^3}$$

$$\frac{\partial z}{\partial x}\cdot\frac{\partial x}{\partial y}\cdot\frac{\partial y}{\partial z}=-\frac{F_x'}{F_z'}\cdot\left(-\frac{F_y'}{F_x'}\right)\cdot\left(-\frac{F_z'}{F_y'}\right)=-1$$

$\frac{\partial z}{\partial x}$ 是一个整体，不可简单拆分

雅克比行列式

对于 $F(x,y,u,v)$ $H(x,y,u,v)$

$$\frac{D(F,H)}{D(u,v)} = \begin{vmatrix} \frac{\partial F}{\partial u} & \frac{\partial F}{\partial v} \\ \frac{\partial H}{\partial u} & \frac{\partial H}{\partial v} \end{vmatrix} = \frac{\partial F}{\partial u}\cdot\frac{\partial H}{\partial v} - \frac{\partial F}{\partial v}\cdot\frac{\partial H}{\partial u}$$

反推

$$\frac{\partial u}{\partial x} = -\frac{\frac{D(F,H)}{D(x,v)}}{\frac{D(F,H)}{D(u,v)}} \qquad \frac{\partial u}{\partial y} = -\frac{\frac{D(F,H)}{D(y,v)}}{\frac{D(F,H)}{D(u,v)}}$$

$$\frac{\partial v}{\partial x} = -\frac{\frac{D(F,H)}{D(u,x)}}{\frac{D(F,H)}{D(u,v)}} \qquad \frac{\partial v}{\partial y} = -\frac{\frac{D(F,H)}{D(u,y)}}{\frac{D(F,H)}{D(u,v)}}$$

例: $\begin{cases} x^2+y^2+uv=1 \\ xy+u^2+v^2=1 \end{cases}$ $P_0(x_0,y_0,u_0,v_0)=(1,0,1,0)$ 求 $\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial v}{\partial x}, \frac{\partial v}{\partial y}$.

$$\left.\frac{D(F,H)}{D(u,v)}\right|_{(1,0,1,0)} = \left.\begin{vmatrix} F'_u & F'_v \\ H'_u & H'_v \end{vmatrix}\right|_{(1,0,1,0)} = \left.\begin{vmatrix} v & u \\ 2u & 2v \end{vmatrix}\right|_{(1,0,1,0)} = -2$$

$$\frac{\partial u}{\partial x} = -\frac{\frac{D(F,H)}{D(x,v)}}{\frac{D(F,H)}{D(u,v)}} = -\frac{\begin{vmatrix} 2x & u \\ y & 2v \end{vmatrix}}{\begin{vmatrix} v & u \\ 2u & 2v \end{vmatrix}} = -\frac{4xv-yu}{2v^2-2u^2}$$

$$\frac{\partial u}{\partial y} = -\frac{\frac{D(F,H)}{D(y,v)}}{\frac{D(F,H)}{D(u,v)}} = \frac{\begin{vmatrix} 2y & u \\ x & 2v \end{vmatrix}}{\begin{vmatrix} v & u \\ 2u & 2v \end{vmatrix}} = -\frac{4yv-xu}{2v^2-2u^2}$$

$$\frac{\partial v}{\partial x} = -\frac{\frac{D(F,H)}{D(u,x)}}{\frac{D(F,H)}{D(u,v)}} = -\frac{\begin{vmatrix} u & 2x \\ 2u & y \end{vmatrix}}{\begin{vmatrix} v & u \\ 2u & 2v \end{vmatrix}} = -\frac{yv-4xu}{2v^2-2u^2}$$

$$\frac{\partial v}{\partial y} = -\frac{\frac{D(F,H)}{D(u,y)}}{\frac{D(F,H)}{D(u,v)}} = -\frac{\begin{vmatrix} v & 2y \\ 2u & x \end{vmatrix}}{\begin{vmatrix} v & u \\ 2u & 2v \end{vmatrix}} = -\frac{xv-4yu}{2v^2-2u^2}$$